

DEPARTMENT OF CSE, CS&IT, AI&DS
COURSE CODE: 24SDCS02 / 24SDCS02E / 24SDCS02P / 24SDCS02L FULL
STACK APPLICATION DEVELOPMENT

#SKILL - 6 Spring MVC Web Request Handling Demo

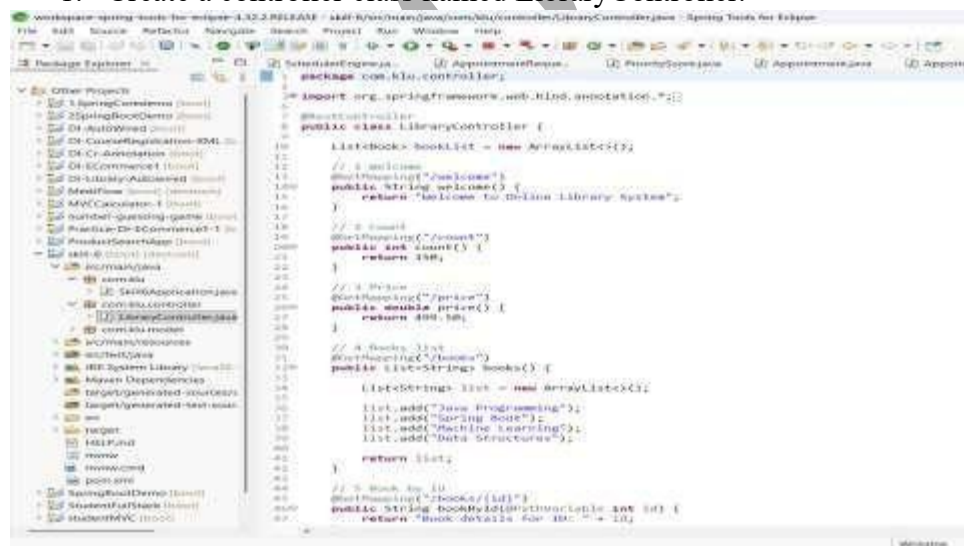
Prerequisites:

- General Idea on Spring Web MVC and Form Handling
- Basic understanding of Spring Boot
- Understanding of HTTP methods (GET/POST)

An online library system wants to allow users to explore books, view specific book details, submit feedback, and interact with the system through various types of requests. To support these features, the backend team needs to implement multiple Spring MVC endpoints demonstrating request mappings, path variables, request parameters, and JSON data handling. Your task is to create a Spring Boot MVC controller that exposes different demo operations using `@RestController`, `@GetMapping`, `@PostMapping`, and `@PathVariable`.

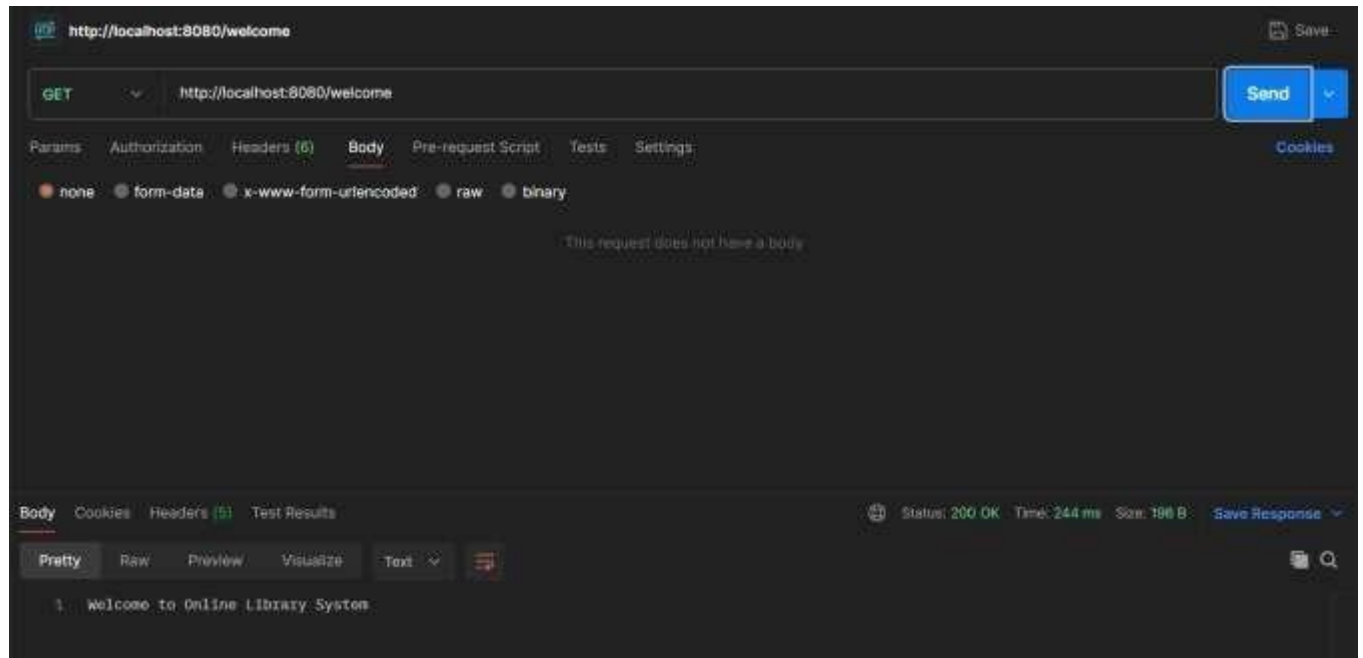
Tasks:

1. Create a controller class named `LibraryController`.

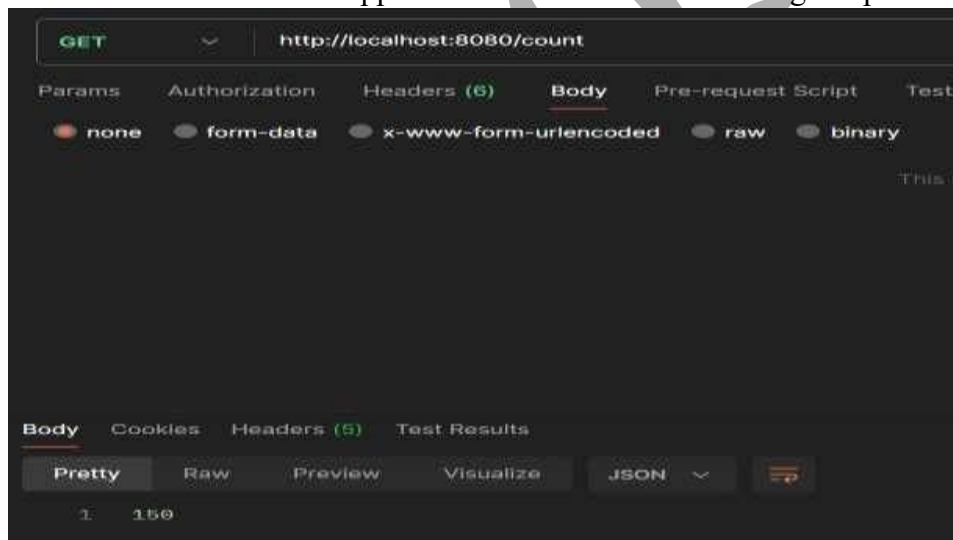


2. Create a method mapped to `/welcome` that returns a welcome message.

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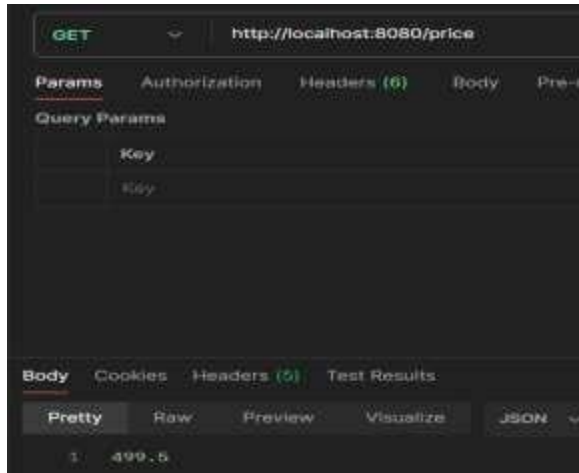


3. Create a method mapped to /count that returns an integer representing total books.

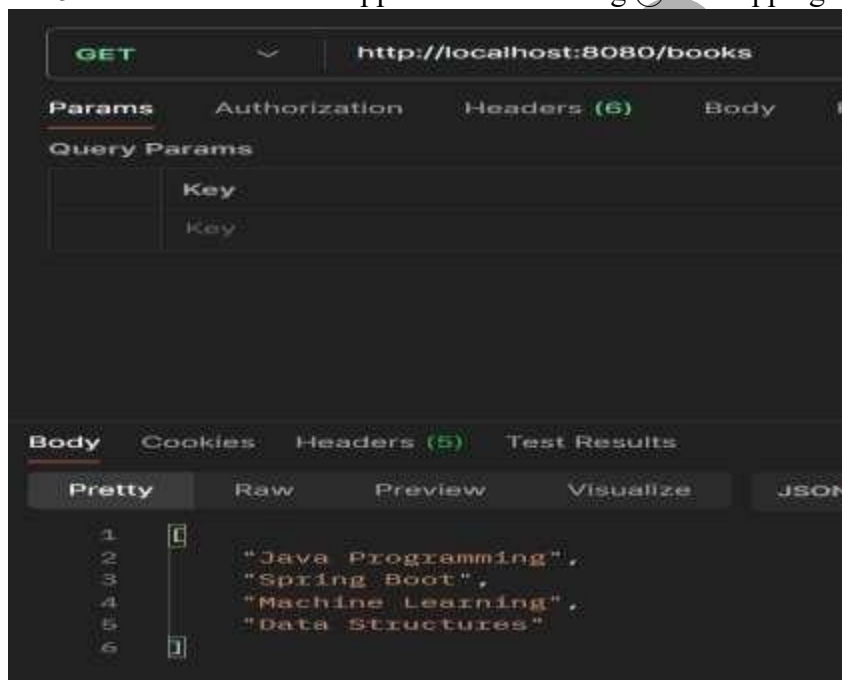


4. Create a method mapped to /price that returns a sample book price as double.

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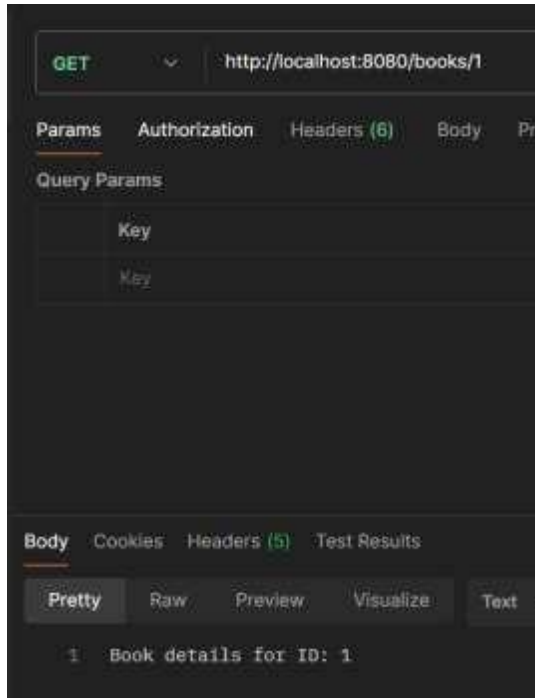


5. Create a method mapped to /books using @GetMapping that returns a list of book titles.

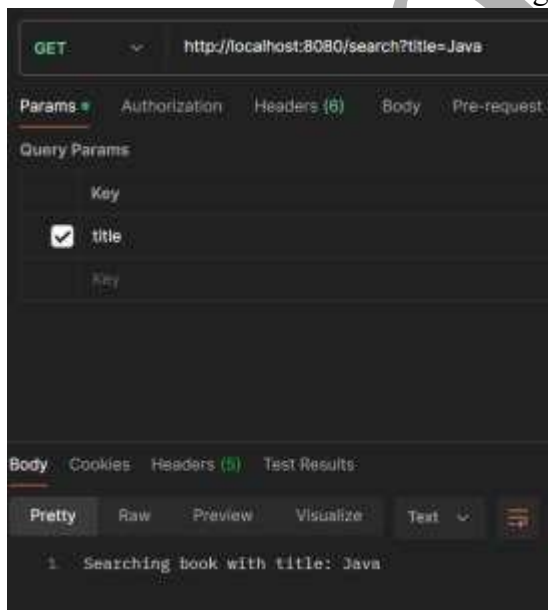


6. Create a method mapped to /books/{id} using @GetMapping and return book details using @PathVariable.

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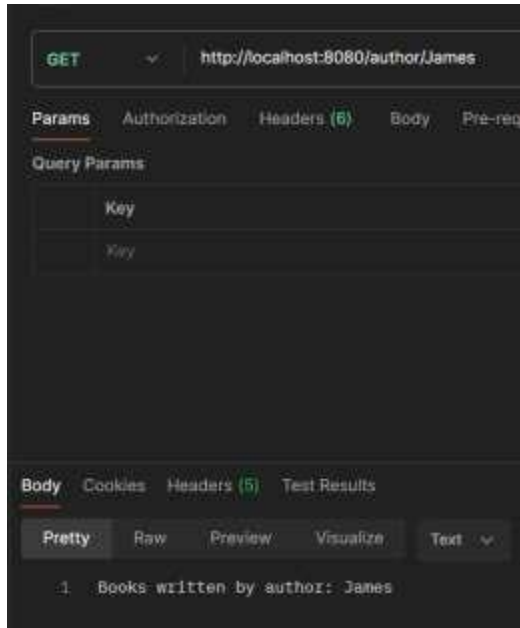


7. Create a method mapped to /search using @GetMapping that accepts a request parameter (title) and returns a confirmation message.

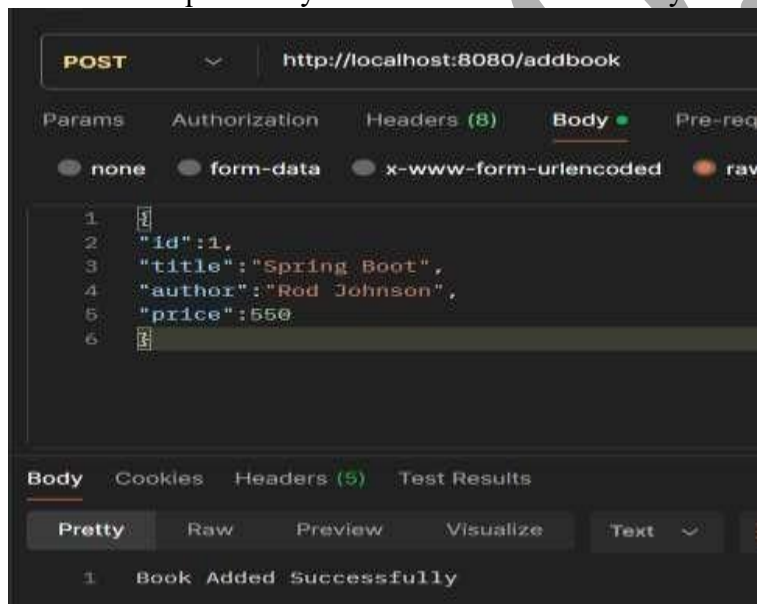


8. Create a method mapped to /author/{name} using @GetMapping that returns a formatted message with the author's name.

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9. Create a method mapped to /addbook using @PostMapping that accepts a Book object from the request body and adds it to an in-memory list.



10. Push the Spring Boot project to GitHub.



1What is the role of @RestController in Spring MVC?

1. What is the difference between @GetMapping and @PostMapping?

@PostMapping handles HTTP **POST** requests and is used to **send or create new data on the server**.

2. How does `@PathVariable` help in handling dynamic URLs?

`@PathVariable` is used to extract values from the **URL path** and pass them as parameters to a controller method.

3. When do we use `@RequestParam`?

`@RequestParam` is used to get **query parameters from the URL** or form data in a request.

4. Can one controller contain multiple request-mapped methods?

Yes, a single controller can contain **multiple request-mapped methods**, each handling different URLs or HTTP request types.

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(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ EMP ID & Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
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